

**UNITED STATES
SECURITIES AND EXCHANGE COMMISSION**
Washington, D.C. 20549

FORM 10-K

(Mark One)

- ☒ **ANNUAL REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**
For the fiscal year ended December 31, 2021
OR
☐ **TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**
For the transition period from _____ to _____.
Commission file number: **001-37580**

Alphabet Inc.

(Exact name of registrant as specified in its charter)

Delaware **61-1767919**
(State or other jurisdiction of incorporation or organization) (I.R.S. Employer Identification No.)

1600 Amphitheatre Parkway
Mountain View, CA 94043
(Address of principal executive offices, including zip code)
(650) 253-0000
(Registrant's telephone number, including area code)

Securities registered pursuant to Section 12(b) of the Act:

<u>Title of each class</u>	<u>Trading Symbol(s)</u>	<u>Name of each exchange on which registered</u>
Class A Common Stock, \$0.001 par value	GOOGL	Nasdaq Stock Market LLC (Nasdaq Global Select Market)
Class C Capital Stock, \$0.001 par value	GOOG	Nasdaq Stock Market LLC (Nasdaq Global Select Market)

Securities registered pursuant to Section 12(g) of the Act:

Title of each class
None

Indicate by check mark if the registrant is a well-known seasoned issuer, as defined in Rule 405 of the Securities Act. Yes ☒ No ☐

Indicate by check mark if the registrant is not required to file reports pursuant to Section 13 or Section 15(d) of the Act. Yes ☐ No ☒

Indicate by check mark whether the registrant (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days. Yes ☒ No ☐

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). Yes ☒ No ☐

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company, or an emerging growth company. See the definitions of "large accelerated filer," "accelerated filer," "smaller reporting company," and "emerging growth company" in Rule 12b-2 of the Exchange Act.

Large accelerated filer	☒	Accelerated filer	☐
Non-accelerated filer	☐	Smaller reporting company	☐
Emerging growth company	☐		

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Indicate by check mark whether the registrant has filed a report on and attestation to its management's assessment of the effectiveness of its internal control over financial reporting under Section 404(b) of the Sarbanes-Oxley Act (15 U.S.C.7262(b)) by the registered public accounting firm that prepared or issued its audit report. ☒

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Exchange Act). Yes ☐ No ☒

As of June 30, 2021, the aggregate market value of shares held by non-affiliates of the registrant (based upon the closing sale prices of such shares on the Nasdaq Global Select Market on June 30, 2021) was approximately \$1,451.1 billion. For purposes of calculating the aggregate market value of shares held by non-affiliates, we have assumed that all outstanding shares are held by non-affiliates, except for shares held by each of our executive officers, directors and 5% or greater stockholders. In the case of 5% or greater stockholders, we have not deemed such stockholders to be affiliates unless there are facts and circumstances which would indicate that such stockholders exercise any control over our company, or unless they hold 10% or more of our outstanding common stock. These assumptions should not be deemed to constitute an admission that all executive officers, directors and 5% or greater stockholders are, in fact, affiliates of our company, or that there are not other persons who may be deemed to be affiliates of our company. Further information concerning shareholdings of our officers, directors and principal stockholders is included or incorporated by reference in Part III, Item 12 of this Annual Report on Form 10-K.

As of January 25, 2022, there were 300,754,904 shares of the registrant's Class A common stock outstanding, 44,576,938 shares of the registrant's Class B common stock outstanding, and 315,639,479 shares of the registrant's Class C capital stock outstanding.

DOCUMENTS INCORPORATED BY REFERENCE

Portions of the registrant's Proxy Statement for the 2022 Annual Meeting of Stockholders are incorporated herein by reference in Part III of this Annual Report on Form 10-K to the extent stated herein. Such proxy statement will be filed with the Securities and Exchange Commission within 120 days of the registrant's fiscal year ended December 31, 2021.

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Alphabet Inc.

Alphabet Inc.
Form 10-K
For the Fiscal Year Ended December 31, 2021

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NOTE ABOUT FORWARD-LOOKING STATEMENTS

This Annual Report on Form 10-K contains forward-looking statements within the meaning of the Private Securities Litigation Reform Act of 1995. These include, among other things, statements regarding:

- the ongoing effect of the novel coronavirus pandemic ("COVID-19"), including its macroeconomic effects on our business, operations, and financial results;
- the growth of our business and revenues and our expectations about the factors that influence our success and trends in our business;
- fluctuations in our revenue growth rate and operating margin and various factors contributing to such fluctuations;
- our expectation that the continuing shift from an offline to online world will continue to benefit our business;
- our expectation that the portion of our revenues that we derive from non-advertising revenues will continue to increase and may affect our margins;
- our expectation that our traffic acquisition costs (TAC) and the associated TAC rate will fluctuate, which could affect our overall margins;
- our expectation that our monetization trends will fluctuate, which could affect our revenues and margins;
- fluctuations in our revenue growth, as well as the change in paid clicks and cost-per-click and the change in impressions and cost-per-impression, and various factors contributing to such fluctuations;
- our expectation that we will continue to periodically review, refine, and update our methodologies for monitoring, gathering, and counting the number of paid clicks and impressions;
- our expectation that our results will be affected by our performance in international markets as users in developing economies increasingly come online;
- our expectation that our foreign exchange risk management program will not fully offset our net exposure to fluctuations in foreign currency exchange rates;
- the expected variability of gains and losses related to hedging activities under our foreign exchange risk management program;
- the amount and timing of revenue recognition from customer contracts with commitments for performance obligations, including our estimate of the remaining amount of commitments and when we expect to recognize revenue;
- fluctuations in our capital expenditures;
- our plans to continue to invest in new businesses, products, services and technologies, systems, land and buildings for data centers and offices, and infrastructure, as well as to continue to invest in acquisitions and strategic investments;
- our pace of hiring and our plans to provide competitive compensation programs;
- our expectation that our cost of revenues, research and development (R&D) expenses, sales and marketing expenses, and general and administrative expenses may increase in amount and/or may increase as a percentage of revenues and may be affected by a number of factors;
- estimates of our future compensation expenses;
- our expectation that our other income (expense), net (OI&E), will fluctuate in the future, as it is largely driven by market dynamics;
- fluctuations in our effective tax rate;
- seasonal fluctuations in internet usage and advertiser expenditures, underlying business trends such as traditional retail seasonality, which are likely to cause fluctuations in our quarterly results;
- the sufficiency of our sources of funding;
- our potential exposure in connection with new and pending investigations, proceedings, and other contingencies;

- the sufficiency and timing of our proposed remedies in response to decisions from the European Commission (EC) and other regulators and governmental entities;
- our expectations regarding the timing, design, and ongoing phased implementation of our new global enterprise resource planning (ERP) system;
- the expected timing, amount, and effect of Alphabet Inc.'s share repurchases;
- our long-term sustainability and diversity goals;

as well as other statements regarding our future operations, financial condition and prospects, and business strategies. Forward-looking statements may appear throughout this report and other documents we file with the Securities and Exchange Commission (SEC), including without limitation, the following sections: Part I, Item 1 "Business;" Part I, Item 1A "Risk Factors;" and Part II, Item 7 "Management's Discussion and Analysis of Financial Condition and Results of Operations." Forward-looking statements generally can be identified by words such as "anticipates," "believes," "estimates," "expects," "intends," "plans," "predicts," "projects," "will be," "will continue," "may," "could," "will likely result," and similar expressions. These forward-looking statements are based on current expectations and assumptions that are subject to risks and uncertainties, which could cause our actual results to differ materially from those reflected in the forward-looking statements. Factors that could cause or contribute to such differences include, but are not limited to, those discussed in this Annual Report on Form 10-K, and in particular, the risks discussed in Part I, Item 1A, "Risk Factors" of this report and those discussed in other documents we file with the SEC. We undertake no obligation to revise or publicly release the results of any revision to these forward-looking statements, except as required by law. Given these risks and uncertainties, readers are cautioned not to place undue reliance on such forward-looking statements.

As used herein, "Alphabet," "the company," "we," "us," "our," and similar terms include Alphabet Inc. and its subsidiaries, unless the context indicates otherwise.

"Alphabet," "Google," and other trademarks of ours appearing in this report are our property. This report contains additional trade names and trademarks of other companies. We do not intend our use or display of other companies' trade names or trademarks to imply an endorsement or sponsorship of us by such companies, or any relationship with any of these companies.

PART I

ITEM 1. BUSINESS

Overview

As our founders Larry and Sergey wrote in the original founders' letter, "Google is not a conventional company. We do not intend to become one." That unconventional spirit has been a driving force throughout our history, inspiring us to tackle big problems and invest in moonshots like artificial intelligence (AI) research and quantum computing. We continue this work under the leadership of Sundar Pichai, who has served as CEO of Google since 2015 and as CEO of Alphabet since 2019.

Alphabet is a collection of businesses — the largest of which is Google. We report Google in two segments, Google Services and Google Cloud; we also report all non-Google businesses collectively as Other Bets. Other Bets include earlier stage technologies that are further afield from our core Google business. We take a long-term view and manage the portfolio of Other Bets with the discipline and rigor needed to deliver long-term returns. Alphabet's structure is about helping each of our businesses prosper through strong leaders and independence.

Access and technology for everyone

The Internet is one of the world's most powerful equalizers; it propels ideas, people and businesses large and small. Our mission to organize the world's information and make it universally accessible and useful is as relevant today as it was when we were founded in 1998. Since then, we have evolved from a company that helps people find answers to a company that also helps people get things done.

We are focused on building an even more helpful Google for everyone, and we aspire to give everyone the tools they need to increase their knowledge, health, happiness, and success. Every year, there are trillions of searches on Google, and 15% of the searches we see every day are new. We continue to invest deeply in AI and other technologies to ensure the most helpful search experience possible. YouTube provides people with entertainment, information, and opportunities to learn something new. And Google Assistant offers the best way to get things done seamlessly across different devices, providing intelligent help throughout a person's day, no matter where they are.

We are continually innovating and building new product features that will help our users, partners, customers, and communities. We have invested more than \$100 billion in R&D over the last five years. In addition, with the onset of

the pandemic, we have focused in particular on features that help people in their daily lives and that support businesses working to serve their customers. For example, we have added live business trends in Google Maps that help users instantly spot when a neighborhood or part of town is near or at its busiest. We have also helped businesses navigate uncertainty during an uneven economic recovery, and we have worked to address the complex challenge of distributing critical information about COVID-19 vaccines to billions of people around the world. Importantly, we have made authoritative content a key focus area across both Google Search and YouTube to help users find trusted public health information.

Other Bets also remain focused on innovation through technology that can positively affect people's lives. For instance, Waymo is working toward our goal of making transportation safer and easier for everyone and Verily is developing tools and platforms to improve health outcomes.

Moonshots

Many companies get comfortable doing what they have always done, making only incremental changes. This incrementalism leads to irrelevance over time, especially in technology, where change tends to be revolutionary, not evolutionary. People thought we were crazy when we acquired YouTube and Android and when we launched Chrome, but those efforts have matured into major platforms for digital video and mobile devices and a safer, popular browser. We continue to look toward the future and to invest for the long term within each of our segments. As we said in the original founders' letter, we will not shy away from high-risk, high-reward projects that we believe in, as they are the key to our long-term success.

The power of AI

Across the company, investments in AI and machine learning are increasingly driving many of our latest innovations and have enabled us to build products that are smarter and more helpful. For example, in May of 2021, we introduced Multitask Unified Model — or MUM — which has the potential to transform how Google helps with complex tasks. MUM is trained across 75 different languages, which means that it can learn from sources written in one language and help bring that information to people in another. It is also multimodal, so it understands information across text and images and, in the future, can expand to more modalities like video and audio. We are currently experimenting with MUM's capabilities to make searching more natural and intuitive and even enable entirely new ways to search.

DeepMind also made a significant AI-powered breakthrough, solving a 50-year-old protein folding challenge, which will help the world better understand one of life's fundamental building blocks, and will enable researchers to tackle new and difficult problems, from fighting diseases to environmental sustainability. DeepMind has since shared its new AlphaFold protein structure database, which doubled the number of high-accuracy human protein structures available to researchers.

Google

For reporting purposes, Google comprises two segments: Google Services and Google Cloud.

Google Services

Serving our users

We have always been a company committed to building helpful products that can improve the lives of millions of people. Our product innovations have made our services widely used, and our brand one of the most recognized in the world. Google Services' core products and platforms include ads, Android, Chrome, hardware, Gmail, Google Drive, Google Maps, Google Photos, Google Play, Search, and YouTube, each with broad and growing adoption by users around the world.

Our products and services have come a long way since the company was founded more than two decades ago. Rather than the ten blue links in our early search results, users can now get direct answers to their questions using their computer, mobile device, or their own voice, making it quicker, easier and more natural to find what they are looking for.

This drive to make information more accessible and helpful has led us over the years to improve the discovery and creation of digital content both on the web and through platforms like Google Play and YouTube. With the continued adoption of mobile, people are consuming more digital content by watching more videos, playing more games, listening to more music, reading more books, and using more apps than ever before. Working with content creators and partners, we continue to build new ways for people around the world to find great digital content.

Fueling all of these great digital experiences are extraordinary platforms and hardware. That is why we continue to invest in platforms like our Android mobile operating system, Chrome browser, and Chrome operating system, as

well as growing our family of hardware devices. We see tremendous potential for devices to be helpful and make people's lives easier by combining the best of our AI, software and hardware. This potential is reflected in our latest generation of hardware products such as Pixel 5a 5G and Pixel 6 phones, the Fitbit Charge 5, Chromecast with Google TV, and the new Google Nest Cams and Nest Doorbell. Creating products that people rely on every day is a journey that we are investing in for the long run.

The key to building helpful products for users is our commitment to privacy, security, and user choice. We protect user privacy and security with products that are secure by default and private by design, and that keep users in control of their data. Our privacy-preserving technologies safeguard individual privacy and enhance data protection. As the Internet evolves, so does our approach to privacy and security. We continue to enhance our anti-malware features in Chrome and drive improvements such as auto-delete controls that automatically delete web and app searches after 18 months. And we continue to keep users and their passwords safe through advances like our built-in password manager.

How we make money

We have built world-class advertising technologies for advertisers, agencies, and publishers to power their digital marketing businesses. Our advertising solutions help millions of companies grow their businesses through our wide range of products across devices and formats, and we aim to ensure positive user experiences by serving the right ads at the right time and by building deep partnerships with brands and agencies.

Google Services generates revenues primarily by delivering both performance and brand advertising that appears on Google Search & other properties, YouTube and Google Network partners' properties ("Google Network properties"). We continue to invest in both performance and brand advertising and seek to improve the measurability of advertising so advertisers understand the effectiveness of their campaigns.

- **Performance advertising** creates and delivers relevant ads that users will click on, leading to direct engagement with advertisers. Performance advertising lets our advertisers connect with users while driving measurable results. Our ads tools allow performance advertisers to create simple text-based ads.
- **Brand advertising** helps enhance users' awareness of and affinity for advertisers' products and services, through videos, text, images, and other interactive ads that run across various devices. We help brand advertisers deliver digital videos and other types of ads to specific audiences for their brand-building marketing campaigns.

We have allocated substantial resources to stopping bad advertising practices and protecting users on the web. We focus on creating the best advertising experiences for our users and advertisers in many ways, including filtering out invalid traffic, removing billions of bad ads from our systems every year, and closely monitoring the sites, apps, and videos where ads appear and blocklisting them when necessary to ensure that ads do not fund bad content.

We continue to look to the future and are making long-term investments that we expect to grow revenues beyond advertising, including revenues from Google Play, hardware, and YouTube non-advertising.

- **Google Play** generates revenues from sales of apps and in-app purchases and digital content sold in the Google Play store.
- **Hardware** generates revenues from sales of Fitbit wearable devices, Google Nest home products, Pixel phones, and other devices.
- **YouTube non-advertising** generates revenues from YouTube Premium and YouTube TV subscriptions and other services.

Google Cloud

Google was a company built in the cloud. We continue to invest in infrastructure, security, data management, analytics, and AI. We see significant opportunity in helping businesses utilize these strengths with features like data migration, modern development environments, and machine learning tools to provide enterprise-ready cloud services, including Google Cloud Platform and Google Workspace. Google Cloud Platform enables developers to build, test, and deploy applications on its highly scalable and reliable infrastructure. Google Workspace collaboration tools — which include apps like Gmail, Docs, Drive, Calendar, Meet and more — are designed with real-time collaboration and machine intelligence to help people work smarter. Because more and more of today's digital experiences are being built in the cloud, Google Cloud products help businesses of all sizes take advantage of the latest technology advances to operate more efficiently.

- **Google Cloud Platform** generates revenues from infrastructure, platform and other services.

- **Google Workspace** generates revenues from cloud-based collaboration tools for enterprises, such as Gmail, Docs, Drive, Calendar and Meet.

Our cloud services are generally provided on either a consumption or subscription basis and may have contract terms longer than a year.

Other Bets

Across Alphabet, we are also using technology to try to solve big problems that affect a wide variety of industries. Alphabet's investment in the portfolio of Other Bets includes emerging businesses at various stages of development, ranging from those in the R&D phase to those that are in the beginning stages of commercialization, and our goal is for them to become thriving, successful businesses in the medium to long term. While these early-stage businesses naturally come with considerable uncertainty, some of them are already generating revenue and making important strides in their industries. Revenues from Other Bets are generated primarily from the sale of health technology and internet services.

Other Bets operate as independent companies and some of them have their own boards with independent members and outside investors. We are investing in the portfolio of Other Bets and being very deliberate about the focus, scale, and pace of investments.

Competition

Our business is characterized by rapid change as well as new and disruptive technologies. We face formidable competition in every aspect of our business, including from:

- General purpose search engines and information services, such as Baidu, Microsoft's Bing, Naver, Seznam, Yahoo, and Yandex.
- Vertical search engines and e-commerce providers, such as Amazon and eBay (e-commerce), Booking's Kayak (travel queries), Microsoft's LinkedIn (job queries), and WebMD (health queries). Some users will navigate directly to such content, websites, and apps rather than go through Google.
- Social networks offered by ByteDance, Meta, Snap, and Twitter. Some users increasingly rely on social networks for p